

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 10: Aircraft Fuel Systems (Tanks, Pumps & Jettison)

Syllabus Source: CAE Oxford (pp. 309-338) | Pass Target: >= 90% AviationExam

1. AVIATION TURBINE FUELS: JET A-1 VS JET A VS JET B

Jet engines burn kerosene-type aviation fuels possessing high energy density, low freezing points, and clean combustion characteristics:

Fuel Designation	Chemical Base & Composition	Freezing Point & Flash Point Specification
Jet A-1	Civil standard kerosene used worldwide (except domestic USA).	Freezing Point: -47°C maximum. Flash Point: +38°C (100°F) minimum. Nominal Density: 0.80 kg/litre (6.7 lb/US Gal).
Jet A	Civil kerosene used primarily in the domestic United States.	Freezing Point: -40°C maximum (7°C higher than Jet A-1!). Flash Point: +38°C minimum. Density: 0.80 kg/litre.
Jet B (Wide-Cut)	Blend of gasoline and kerosene (military equivalent JP-4).	Freezing Point: -60°C maximum. Flash Point: VERY LOW (-20°C). Highly volatile and hazardous; used only in extreme Arctic/polar operations.

[!] EXAM TRAP (AVIATIONEXAM) - Fuel Temperature & Freezing Monitoring in Long-Range Cruise

- Minimum Fuel Temperature Limit: Fuel temperature in flight must not be allowed to drop closer than 3°C above the fuel freezing point (e.g. For Jet A-1 with freezing point -47°C, minimum allowable fuel temp is -44°C)! - Corrective Actions for Cold Fuel: 1. Accelerate to higher Mach number (increases kinetic Ram rise heating: $TAT = SAT \times (1 + 0.2 M^2)$); 2. Descend to a lower, warmer altitude.

2. TANK ARCHITECTURE, BOOST PUMPS & CROSSFEED

Commercial airliners store fuel primarily in the wings (wet wing integral tanks) and center fuselage tank:

- **Integral Tanks (Wet Wing):** The wing structure itself (spars, skin, ribs) forms the fuel tank, sealed with polysulfide sealant. Saves massive weight compared to separate fuel cells.
- **Wing Bending Relief:** Fuel in the wings counteracts upward aerodynamic wing bending moments during flight, reducing stress on the wing root! For this reason, fuel from the **CENTER TANK** is always burned **FIRST**, keeping wing fuel loaded until later in the flight.
- **Submerged Fuel Boost Pumps:** Each tank contains AC motor-driven submerged centrifugal pumps. Center tank pumps have higher output pressure (override/jettison pumps) than wing pumps, ensuring center tank fuel feeds engines preferentially.
- **Crossfeed Valve:** Allows any fuel tank to feed any engine, permitting lateral fuel balance corrections in flight, or allowing remaining engines to feed from all tanks during engine-out operations.
- **Fuel Jettison (Dumping):** Mandatory on aircraft whose Maximum Take-Off Mass (MTOM) significantly exceeds Maximum Landing Mass (MLM). Must be capable of dumping fuel rapidly down to maximum landing mass within 15 minutes, while maintaining sufficient fuel reserve to fly to an alternate.

3. AIRCRAFT REFUELING SAFETY & NITROGEN INERTING (OBIGGS)

Ground refueling presents extreme fire and static discharge hazards:

- Electrical Bonding: Before fuel hose connection, a grounding/bonding wire must be attached between the fuel truck and aircraft to equalize electrostatic potential and prevent sparks.
- Nitrogen Inerting (OBIGGS): On-Board Inert Gas Generating Systems separate nitrogen from bleed air and pump nitrogen-enriched air (NEA, oxygen < 9%) into fuel tank vapor ullage spaces, eliminating flammable oxygen mixtures (preventing TWA 800 center tank explosion scenario).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Why is center tank fuel burned before wing tank fuel during normal flight in transport aircraft?

[A] Because center tank fuel freezes faster [B] To preserve fuel mass in the wings as long as possible, providing upward wing-bending relief against aerodynamic lift forces [C] Because center tank fuel cannot be jettisoned [D] To keep the center of gravity as far aft as possible

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The weight of fuel stored inside the wings opposes upward aerodynamic lift, reducing bending moments at the wing root. Consuming center tank fuel first preserves this structural wing-bending relief.

Q2: What is the specified maximum freezing point of standard international Jet A-1 fuel?

[A] -40°C [B] -47°C [C] -50°C [D] -60°C

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Jet A-1 has a mandatory maximum freezing point of -47°C. (Jet A, used in the US, has a freezing point of -40°C).

Q3: In a long-range jet flying in polar airspace, fuel temperature indicates -43°C while burning Jet A-1 (freezing

[A] Jettison the cold fuel immediately [B] Descend to a warmer altitude or increase Mach number to increase aerodynamic kinetic heating [C] Turn off the fuel boost pumps [D] Shut down one engine to heat the wing

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Fuel temp must remain at least 3°C above freezing point (minimum -44°C for Jet A-1). At -43°C, fuel is dangerously close to waxing. Pilots must accelerate (higher Mach = higher TAT ram rise) or descend into warmer air.